

Shivashriganesh P. Mahato

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BIOGRAPHY

I am a third-year PhD student at the University of Pennsylvania, advised by Prof. Eva L. Dyer, interested in representation learning from neural data. My work spans invasive (intracranial EEG, neural spikes, calcium imaging) and non-invasive (EEG, EMG) neural modalities as I develop new approaches for learning how to decode from brain signals. Of particular interest to me is the challenge of integrating methods across sources of heterogeneity, including temporal drift, subject-to-subject variability, and diverse behaviors. I am interested in developing flexible tools that can enable clinical outcomes downstream, empower assistive technologies, and hopefully enhance our understanding of the brain.

EDUCATION

PhD in Computer and Information Science

Aug 2024 – Present

University of Pennsylvania (transferred from Georgia Tech) • Advisor: Prof. Eva L. Dyer

- Relevant coursework: ML for Neural/Behavioral Data, Systems Neuroscience, Discrete Generative Models

MS in Computer Science

Aug 2022 – Aug 2023

Texas A&M University • Advisor: Prof. Yoonsuck Choe

- Relevant coursework: Theory of Deep Learning, ML, Operating Systems, Info Retrieval, Stochastic Processes
- Thesis: *Comparison of Evolved Memory Mechanisms in Neural Networks*

BS in Applied Mathematics

Aug 2019 – May 2022

Texas A&M University

- Minors in Computer Science and Statistics, graduated Summa Cum Laude with Honors
- Relevant coursework: Topological Data Analysis, Multivariate Analysis, Statistical Learning, Applied Analytics

PUBLICATIONS

- Andre, A.*, **Mahato, S. P.***, Arora, V., et al. "BrainWideBench: Benchmarking Large-Scale Pretraining and Across-Animal Transfer in Multi-Region Neural Recordings." Under review at *2026 Neural Information Processing Systems (NeurIPS)*.
- **Mahato, S.P.***, Xiao, J.*, Andre, A., et al. (2025). [A scalable self-supervised method for modeling human intracranial recordings during natural behavior](#). *NeurIPS 2025 Workshop on Foundation Models for the Brain and Body*. Spotlight (top 15%).
- Arora, V.*, Lachi, D.*, **Mahato, S.P.**, Azabou, M., Chen, Z., Dyer, E.L. (2025). [Exploiting All Laplacian Eigenvectors for Node Classification with Graph Transformers](#). *NeurIPS 2025 Workshop on New Perspectives in Graph Machine Learning*.
- Lachi, D., Mohammadi, M., Meyer, J., Arora, V., **Mahato, S.P.**, Palczewski, T., Dyer, E.L. (2025). [Learning from Multi-Table Relational Data with the Relational Graph Perceiver](#). *EurIPS 2025 Workshop on AI for Tabular Data*.
- **Mahato, S.P.**, Kaliyur, S., Chung, J.R., Choe, Y. (2024). [Use of External Markers by Reactive Agents as an Easier Evolutionary Route Toward Memory](#). *2024 International Joint Conference on Neural Networks (IJCNN)* (pp. 1–9). IEEE.

RESEARCH EXPERIENCE

Graduate Research Assistant (PhD), Neural Data Science (NerDS) Lab

Aug 2024 – Present

PI: Prof. Eva L. Dyer • University of Pennsylvania

- Created and deployed a new benchmark for neuro-foundation models on the IBL Brainwide Map. We setup an evaluation suite across three task domains (decoding, neural activity prediction, and anatomical localization) for large-scale learning from neural spiking activity. We rigorously compared classical models and state-of-the-art foundation models to serve as standard baselines for the community (in review, NeurIPS 2026).
- Developed a self-supervised learning framework for large-scale intracranial EEG, which reconstructs masked channels to learn general neural representations that improve downstream decoding performance (NeurIPS Workshop 2025).
- Co-authored works on graph learning, including a principled method for frequency-informed selection of Laplacian

eigenvector positional embeddings in Graph Transformers, and large-scale graph learning on dynamic relational data (NeurIPS Workshop 2025).

Graduate Researcher (MS Thesis), Brain Networks Lab

Jan 2023 – Aug 2023

PI: Prof. Yoonsuck Choe • Texas A&M University

- Studied the evolution of memory in neural controllers in an empirical testbed, focusing on how memory could evolve from fully reactive agents and highlighting the feasibility of an external-to-internal route towards memory development to qualify previous works (IJCNN 2024).
 - Under a simulated neuroevolution framework (NEAT), we trained different memory-capable systems (RNN, TDNN, FFN with external interactive capabilities [FFWD]) in a memory-dependent RL task.
 - We then analyzed network topologies, developed a novel characterization of evolutionary efficiency, and qualitatively compared dynamics in-distribution as well as with perturbed environmental conditioning to demonstrate that FFWD is a rudimentary memory mechanism.

REU Scholar, Department of Statistics

Apr 2022 – Aug 2022

PI: Prof. Sharmista Guha • Texas A&M University

- Compared multiple neuroimaging datasets for lab initiative towards modeling clinical disease progression, selecting OASIS-3 for its multimodal imaging and cognitive (Alzheimer's) assessments.
- Developed a preprocessing pipeline (registration of Diffusion Tensor Imaging to T1 space, tractography analysis, generation of adjacency matrices) to obtain input features for Bayesian Network Regression.

REU Scholar, Department of Mathematics

May 2020 – Jul 2020

PI: Prof. Wencai Liu • Texas A&M University

- Contributed to a re-formulation of the standard solution to the Ambarzumian problem with Dirichlet boundary conditions, framing it as an inverse spectral problem.

Student Research Assistant, Texas A&M AgriLife

Oct 2019 – Mar 2020

PI: Prof. Muthukumar Bagavathiannan • Texas A&M University

- Performed extensive data annotation on crop field imagery captured by UAV, yielding a large-scale supervised dataset with segmentation labels.
- Tested various feature extraction schemes and trained a preliminary model (finetuned U-Net-ResNet50) for semantic segmentation of weeds and crops, achieving 90% pixel-wise classification accuracy.

PROFESSIONAL EXPERIENCE

Capital One

Chicago, IL

Senior Associate Software Engineer

Jul 2024 – Aug 2024

- Enhanced observability of universal payments infrastructure through end-to-end traceability instrumentation and improved metrics collection and logging.

Associate Software Engineer

Aug 2023 – Jul 2024

- Led design, implementation, and full-stack testing (E2E, performance, resiliency) of a major payments scheduling feature; established team-wide performance testing standards as SME.
- Architected and deployed an automated email reporting service for payments fulfillment reconciliation, customized per line of business.
- Authored AWS cloud compliance procedures and on-call troubleshooting runbooks, codifying operational knowledge for team-wide use.

Sellers Shield

Austin, TX

Software Engineer II

Jan 2023 – Mar 2023

- Standardized backend testing infrastructure and led OAuth implementation for the partner-facing API; implemented UI wireframes for platform migration.

Capital One

Plano, TX

Software Engineering Intern

Jun 2022 – Aug 2022

- Built and deployed a serverless AWS web application for audit log validation during platform migration, cutting the validation process from days to hours while eliminating downtime costs.

Ascension Digital Studio

Austin, TX

Software Engineering Intern

May 2021 – Aug 2021

- Reduced test time by ~45% by embedding a Mongo database into the test suite; developed and shipped features for a surgical scheduling system with production deployments on GCP.
- Provisioned the surgical scheduling manager on the enterprise service layer to enable smooth migration from legacy EHR systems.

Vitalus Health

Katy, TX

Contract Software Developer

Feb 2021 – May 2021

- Implemented a general-purpose SMS API serving as the communication backbone for all customer-facing interactions.
- Built custom logging, a patient scheduling system, and a data provisioning layer supporting rapid migration and NLP chatbot integration.

TEACHING EXPERIENCE

Teaching Assistant, CIS 7000: Deep Learning for Time Series (DL4TS)

Spring 2026

University of Pennsylvania

Teaching Assistant, ECE 4252/8803: Fundamentals of Machine Learning

Fall 2024

Georgia Institute of Technology

CONFERENCE & WORKSHOP PARTICIPATION

- Presented “BrainWideBench: Benchmarking large-scale pretraining on multi-region neural recordings” at *ARNI Working Group on Neuro-Foundation Models*. New York City, NY, USA (February 2026).
- Presented “A scalable self-supervised method for modeling human intracranial recordings during natural behavior” at *NeurIPS 2025 Workshop on Foundation Models for the Brain and Body* (oral and poster). San Diego, CA, USA (December 2025).
- Attended *2025 CCN Workshop: “Functional alignment of information encoded in neural activity”* (Dartmouth). Hanover, NH, USA (September 2025).
- Presented “Large-Scale Deep Learning in Neuroscience” at *Computational Neural-Engineering Training Program Tutorial Series* (Georgia Tech). Atlanta, GA, USA (June 2025).
- Presented “Machine Learning for Adaptive DBS” at *Adaptive Deep Brain Stimulation Workshop* (University of Pennsylvania). Philadelphia, PA, USA (April 2025).
- Presented “Large-Scale Deep Learning in Neuroscience” at *Data-Driven and Large-Scale Modeling in Neuroscience Workshop* (Georgia Tech). Atlanta, GA, USA (April 2025).
- TAed “Transformers in Neuroscience” Tutorial at *Computational and Systems Neuroscience (COSYNE)*. Montreal, Canada (March 2025).
- Presented “Use of External Markers by Reactive Agents as an Easier Evolutionary Route Toward Memory” at *IEEE World Conference on Computational Intelligence (WCCI)*. Yokohama, Japan (June 2024).

HONORS, AWARDS & CERTIFICATIONS

- President's Fellowship, Georgia Institute of Technology (2024)
- IEEE Computational Intelligence Society (CIS) “Industry Early Career” Participation and Travel Grant (2024)
- AWS Certified Solutions Architect, Associate (May 2024 – May 2027)
- Honors Degree in Mathematics, Texas A&M Department of Mathematics (2022)
- Certificate of Excellence, Texas A&M Department of Mathematics (2022)
- Classroom Excellence Award for MATH 470 (Cryptography), Texas A&M Department of Mathematics (2021)

PROFESSIONAL AFFILIATIONS

- Reviewer, Neural Information Processing Systems (*NeurIPS*) (2026)
- Reviewer, International Conference on Learning Representations (*ICLR*) (2026)
- Reviewer, Computational and Systems Neuroscience (*COSYNE*) (2026)
- Reviewer, NeurIPS 2025 Workshop on Foundation Models for the Brain and Body (2025)
- Reviewer, International Joint Conference on Neural Networks (*IJCNN*) (2024, 2025)

- Reviewer, IEEE Transactions on Cognitive and Developmental Systems (2024, 2025)
- Member, Institute of Electrical and Electronics Engineers (2024 – 2025)
- Member, IEEE Computational Intelligence Society (2024 – 2025)

MENTORSHIP

- Reid Laughton, MS Computer and Information Science, University of Pennsylvania (Ongoing)
- Keshav Balaji, MS Computer and Information Science, University of Pennsylvania (Ongoing)
- Jake Manning, MS Computer Science, Texas A&M University (2025)
- Tanu Shree, MS Computer Science, Texas A&M University (2023)
 - Thesis: *Classification of Cricket Songs Based on Their Genus and Species*

TECHNICAL SKILLS

- **ML Tools:** pytorch, lightning, numpy, cupy, pandas, sklearn, keras, neat-python, wandb, slurm
- **Languages:** Python, Java, C++, Javascript, R, SQL
- **Web Tools:** Spring Boot, Django, NodeJS, Flask, React, VueJS
- **Dev Tools:** Git, Docker, Kubernetes, Linux, Jenkins, Splunk, New Relic, Jira, Terraform
- **Cloud:** AWS, Google Cloud Platform (GCP), Firebase, Cloudflare